

DEVELOPMENT_GUIDE

Development Guide

Prerequisites

- JDK 21
- Maven 3.9+
- Node.js + Angular CLI (frontend)
- PostgreSQL with DB bankone and user bankone_user
- Optional: Open Liberty under ~/tools/wlp (server name bankone)

Backend

```
cd "BankOne/BankOne"  
# Embedded  
mvn spring-boot:run  
# Liberty (recommended for this project)  
./scripts/redeploy-liberty.sh
```

Key config: src/main/resources/application.properties

Property	Purpose
spring.datasource.*	PostgreSQL connection
spring.jpa.hibernate.ddl-auto=update	Schema sync
spring.sql.init.mode=always	Runs schema.sql
server.port=8080	Embedded only
jwt.secret / jwt.expiration	JWT signing

After Liberty redeploy, smoke login expects admin / Admin@123.

Frontend

```
cd "BankOne/BankOne-Frontend"  
npm install # first time  
ng serve
```

Set API host in src/app/core/config/api-config.ts (currently points at Liberty / LAN host on port 9080).

Coding conventions

Backend

- Package by domain: com.bankone.{module}.{controller|service|repository|entity|dto|...}
- Prefer service interface + *Impl
- Search via JPA Specification classes
- Map API responses through DTOs for accounts/policies; customers currently return entity (known smell)

- Security roles: ROLE_ prefix handled by Spring; matchers use ADMIN, EMPLOYEE, MANAGER

Frontend

- Feature folders under src/app/features/
- Shared HTTP in src/app/core/services/
- Models in src/app/core/models/
- Bearer token via auth.interceptor.ts
- Dialogs as standalone components (e.g. opening-deposit-dialog/)

Teach-while-building

Project rule (.cursor/rules/teach-while-building.mdc): unless the user says to apply code, explain snippets for IntelliJ paste. Documentation sync still applies when code lands.

Documentation sync checklist (every change)

1. Update module doc under docs/MODULES/
2. Update API / schema / sequences / call flow if surfaces changed
3. Append docs/CHANGELOG.md
4. Refresh Mermaid diagrams when structure changes

Common pitfalls

Symptom	Likely cause
405 on new GET endpoint	Stale Liberty WAR — redeploy
Opening CURRENT fails on deposit	Policy requires min opening (seed CURRENT/INR = 5000)
CORS errors from phone/LAN	Firewall / AP isolation; CORS already allows 192.168.0.4
Roles missing after login	Roles not in JWT — check user_roles and filter reload

Suggested next build slices

1. Transaction ledger foundation
2. Account detail + debit/credit history
3. Withdraw
4. Account list filters / polish
5. Status transition rules
6. Maker-checker